

Aussie Primary Academy | Australian Curriculum v9.0 | Learn, Investigate and Explain

Name: _____

Date: _____

LEARNING INTENTION

identify forces exerted by objects and investigate effects of frictional, gravitational and magnetic forces on motion

KEY IDEA

A force is a push or pull. Contact friction opposes motion; gravity pulls masses toward Earth; magnetic forces can attract or repel without contact.

WORKED MODEL

A toy car slows sooner on carpet because greater friction opposes its motion; gravity pulls it down a ramp.

SUCCESS CRITERIA - I CAN

[] use key vocabulary [] classify or model accurately [] explain using evidence [] correct a misconception

A. BUILD THE SCIENCE IDEA

1. Define and use these terms: friction and gravity.

2. Match examples to: friction, gravity, magnetic force.

3. Annotate the worked model with two observations and one scientific explanation.

B. APPLY AND EXPLAIN

4. Sort six examples from this context: a rolling ball, parachute, fridge magnet, bicycle brake, falling leaf and compass

5. Explain two classifications using observable evidence, not appearance alone.

6. Identify the changed condition, the effect and one condition that should stay the same.

C. INDEPENDENT CHECK

7. Draw or construct a labelled scientific model that shows the relationship.

8. Write a three-sentence explanation using because, therefore and evidence.

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D. APPLY SCIENCE IN CONTEXT

9. Case study: A toy car travels down the same ramp onto carpet, tile and felt while distance is measured.

10. Record the important observations in a two-column evidence table.

11. Write a claim-evidence-reasoning conclusion and state one limitation.

E. REASON, EVALUATE AND TRANSFER

12. Find and correct the inaccurate statement in the case study; explain why it is inaccurate.

13. Apply the same science idea to a different everyday situation and predict what would happen.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

[] I can identify forces exerted by objects and investigate effects of frictional, gravitational and magnetic forces on motion.

Circle one: independently | with one reminder | with support

Evidence from my work that proves I understand: _____

The most difficult science idea was _____ because _____

One correction or improvement I made: _____

A question I still have: _____

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Accept equivalent responses when the student uses accurate science ideas, relevant evidence and clear reasoning.

COMPLETE ANSWER KEY

1. Friction opposes surfaces moving; gravity attracts masses; magnetic force attracts/repels magnetic materials.
2. Brake/carpet=friction; falling/parachute involve gravity; magnet/compass magnetic.
3. Carpet friction reduces travel; gravity drives motion down ramp.
4. Classifications may involve more than one force if explained.
5. Evidence should describe motion change.
6. Changed: surface; measured: distance; controls include car, ramp height, release and start point.
- 7-8. Force arrows and causal explanation required.
9. Compare repeated distances across surfaces.
10. Evidence table records surface and trials/average.
11. Conclusion identifies pattern; uneven release is a limitation.
12. Forces can act without visible contact; gravity and magnetism demonstrate this.
13. Accept braking, sport, playground or tool application.

TEACHER CHECK AND NEXT STEP

[] Uses accurate vocabulary and a complete cause-and-effect explanation.

[] Supports conclusions with observations or data.

Next step: revisit one misconception, then transfer the idea to a new example.

COMMON MISCONCEPTION / WATCH OUT

A moving object does not need a continuing forward force when forces are balanced; friction often slows it.

Magnets do not attract every metal.

EXTENSION AND RE-TEACH

Re-teach: model one difficult item, complete one together, then ask the student to explain an independent correction.

Extension: apply the science idea to a new real-world example and justify the explanation with observations or data.